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2/30 1994 TRIAL

Question 1

- (a) Find correct to 2 significant figures the value of

$$\sqrt{\frac{7.849 \times (6.27)^2}{19.863}}$$

- (b) Solve
- $3(x-2) = x+4$

- (c) Given
- $A = 2\pi r(r+h)$
- , find
- h
- correct to one decimal place, given
- $A = 105$
- ,
- $r = 3.2$
- .

- (d) Make
- r
- the subject of

$$S = \frac{a}{1-r}$$

- (e) Factorize
- $x^3 - 8$
- .

Question 2

- (a) Evaluate
- $\sin 1.7$
- correct to 2 decimal places.

- (b) Simplify
- $\frac{1 - \cos^2 \theta}{\sin \theta \cos \theta}$

- (c) Fred receives a pay rise of
- $4\frac{1}{2}\%$
- . If his new salary is \$625/week, find his wage before the pay increase.

- (d) Find the exact value, showing all working out:

$$27^{-\frac{1}{3}} \times 9^{\frac{3}{2}}$$

- (e) Express as a single fraction in simplest form.

$$\frac{2x-3}{2} - \frac{x-2}{5}$$

Question 3

- (a) The points A & B have co-ordinates (2,0) & (0,-2) respectively.
- Draw a diagram, clearly marking A & B
 - Find the gradient AB
 - Show the equation of line ℓ , that passes through A & is perpendicular to AB is given by $x + y = 2$.
 - Show that C, the point of intersection of ℓ & the y axis has co-ordinates (0,2).
 - If D is the point (-2, 0), write down the equation of the circle passing through A, B, C, D.
 - Show that the area between the circle ABCD & the quadrilateral ABCD is $4(\pi - 2)$ units².
- (b) If α & β are the roots of the equation $3x^2 - 15x + 7 = 0$, find
- $\alpha\beta$
 - $\frac{1}{\alpha} + \frac{1}{\beta}$.

Question 4.

- (a) Differentiate
- $x^3 + \frac{1}{x^2}$
 - $2xe^{x^2}$
 - $\ln\left(\frac{x}{\cos x}\right)$
- (b) Find $\int \sin 2x \, dx$.
- (c) Evaluate the following definite integrals correct to two decimal places.

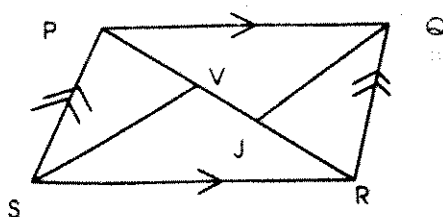
(i) $\int_1^2 \sqrt{3x-2} \, dx$

(ii) $\int_1^e \left(\frac{2}{x} + x\right) dx$

3

Question 5

(a)



PQRS is a parallelogram. TQ bisects angle PQR and VS bisects angle PSR.

- Copy this diagram onto your answer sheet.
 - state why $\angle PQR = \angle PSR$
 - Prove that $\triangle PVS$ & $\triangle RTQ$ are congruent.
 - Hence find the length of TV if $PR = 20$ cm & $TR = 8$ cm.
- (b) The derivative of a function is given by

$$p'(x) = 15(5x - 1)^2$$

If $p(0) = 10$, find the equation of $p(x)$.

Question 6

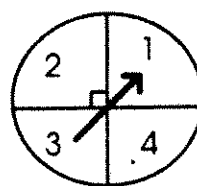
- (a) Consider the function $y = \sqrt{9 - x^2}$

- State its domain & range
- On the number plane shade in the region where the following inequalities hold simultaneously :

$$y < \sqrt{9 - x^2} \text{ and } x \geq 0.$$

- Does the point $(2, -4)$ lie in this region? Use algebra to justify your answer.

- (b) The spinner shown below is used in a game.



It is spun twice & the score recorded after each spin. Find the probability that :

- in each of the two spins the result is 4.
- the sum of the two spins is 4.

- (c) The table below gives the values of $f(t)$ for $0 \leq t \leq 2$.

t	0	0.5	1	1.5	2
f(t)	0	0.30	0.37	0.33	0.27

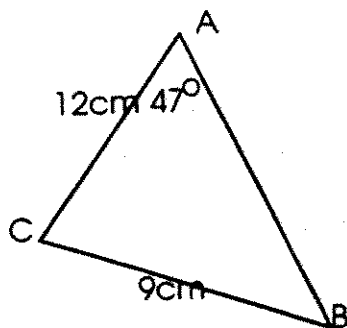
Use the Trapezoidal Rule with 5 function values to evaluate

$$\int_0^2 f(t) dt \text{ correct to one decimal place.}$$

Question 7

(a)

Triangle ABC is an acute triangle in which $AB = 12$ cm, $BC = 9$ cm & angle $BAC = 47^\circ$.



- (i) Find the size of $\angle ACB$, correct to the nearest degree.
 - (ii) Find the area of $\triangle ABC$. Give your answer correct to 2 significant figures.
- (b) The first 3 terms of an AP are 50, 43, 36.
- (i) Write down the formula for the n^{th} term for this sequence.
 - (ii) If the last term of the sequence is -27, how many terms are there in this sequence.
 - iii) Find the sum of the series.
- (c) Find the values of k for which the quadratic equation $x^2 - (k+3)x + (k+6) = 0$ has equal and rational roots.

Question 8

- (a) For the curve $y = (x-2)^2(x+1)$.
- (i) Find the x and y intercepts.
 - (ii) Find the stationary points, if any and determine their nature.
 - (iii) Hence, draw a sketch of the curve in the domain $-2 \leq x \leq 3$.
- (b) Find the equation of the tangent to the curve $y = x^2 + \frac{2}{x} + 4$ at the point $P(-1, 3)$.

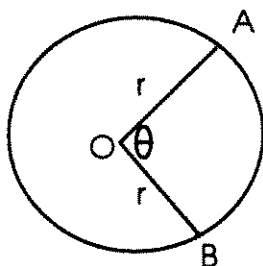
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Question 9

- (a) For the parabola $x^2 = 8y$ find
- the focal length
 - the co-ordinates of the focus
 - the equation of the directrix.
- (b) A stranded motorist on the Nullabor Plains began walking to the nearest watering hole that is located 130 km away. Knowing that she could not cover the entire distance in one day, she planned to walk 75 km the first day & then one third of the previous day's distance each day thereafter. Will she ever reach the watering hole? Justify your answer.
- (c) For the function $f(x) = 2 \cos x$
- State the range of $f(x)$
 - Draw a neat sketch of $f(x) = 2 \cos x$ for $0 \leq x \leq 2\pi$
 - Calculate the area enclosed between the curve $y = 2 \cos x$, the x axis & the lines $x = \frac{\pi}{2}$ & $x = \frac{3\pi}{2}$

Question 10

- (a) In the circle shown arc AB subtends an angle of θ radians at the centre, the radius is r cm., & the perimeter of the sector is 8 cm.



- Show, step by step, that $\theta = \frac{8}{r} - 2$ & the area of the sector is given by $A = 4r - r^2$.
 - Hence, find the radius of the sector of maximum area when the perimeter of the sector AOB is 8 cm.
 - Find the maximum area.
- (b) At the beginning of each year for 25 years, Wally invests \$2300 in a superannuation fund, on which he is paid 12% p.a. interest.

Find the amount his investment is worth at the end of the 25th Year.

End of examination

QUESTION 1

(a) $\sqrt{\frac{784 \times (6.27)^2}{19803}}$
 ≈ 3.94141608 ②
 ≈ 3.9 (to 2 sig fig)

(b) $3(x-2) = x+4$
 $3x-6 = x+4$
 $2x = 10$
 $x = 5$ ②

(c) $A = 2\pi r(h+h)$
 $105 = 2\pi \times 3.2(3.2+h)$
 $\frac{105}{6.4\pi} = 3.2+h$
 $h = \frac{105}{6.4\pi} - 3.2$
 $h \approx 2.02227159$
 $h \approx 2.0$ (1 d.p.) ③

(d) $S = \frac{a}{1-r}$
 $S-Sr = a$
 $Sr = S-a$
 $r = \frac{S-a}{S}$
 $r = 1 - \frac{a}{S}$ ③

(e) $x^3 = 8$
 $x^3 - 2^3$ ②
 $= (x-2)(x^2+2x+4)$

QUESTION 2

(a) $\sin 1.7 \approx 0.9916648$
 ≈ 0.99 (2 d.p.) ①

(b) $\frac{1-\cos^2\theta}{\sin\theta\cos\theta} = \frac{\sin^2\theta}{\sin\theta\cos\theta}$
 $= \frac{\sin\theta}{\cos\theta}$ ③
 $= \tan\theta$

(c) 104.5% of salary = \$62
 $1\% = \frac{62}{104.5}$
 $100\% = \frac{62 \times 100}{104.5}$
 $\approx 59.32\%$

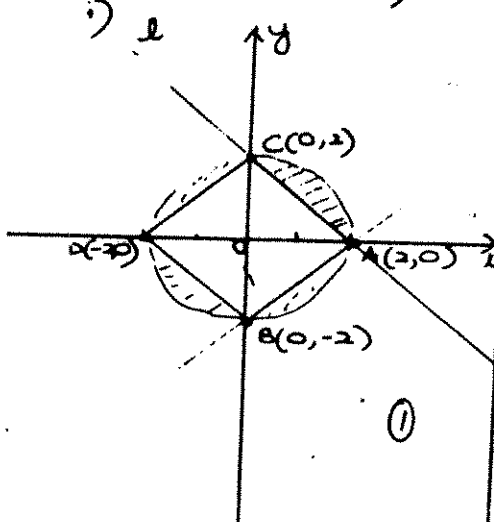
$\therefore$ his wage before the increase was \$59.09 (nearest \$) ③

(d) $27^{-\frac{1}{3}} \times 9^{\frac{1}{3}}$
 $= \frac{1}{\sqrt[3]{27}} \times \sqrt[3]{9}$
 $= \frac{1}{3} \times 27$ ②
 $= 9$

(e) $\frac{2x-3}{2} - \frac{x-2}{5}$
 $= \frac{5(2x-3) - 2(x-2)}{10}$ ③
 $= \frac{10x - 15 - 2x + 4}{10}$
 $= \frac{8x - 11}{10}$

QUESTION 3

(a) $A(2,0)$ $B(0,-2)$



i) $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{-2 - 0}{0 - 2}$
 $m_{AB} = 1$ ①

ii) $2m = -1$ at $A(2,0)$
 $m = \frac{y - y_1}{x - x_1}$
 $-1 = \frac{y - 0}{x - 2}$
 $y = -x + 2$ ②

∴ $x+y=2$ as req'd.

iv) cuts y axis $\Rightarrow x=0$
 $\therefore 0+y=2$
 $y=2$ ①
 $\Rightarrow C(0,2)$ as req'd.

v) circle ABCD,
 centre (0,0) $r=2$
 $x^2 + y^2 = 4$ ①

vi) Req'd area
 $= A \text{ of circle} - A \text{ of quad}$
 $= \pi r^2 - \frac{1}{2}xy$
 $= \pi \times 4 - \frac{1}{2} \times 4 \times 4$
 $= 4\pi - 8$ ②
 $A = 4(\pi - 2) \text{ u}^2$ as req'd.

(b) $3x^2 - 15x + 7 = 0$
 $a=3$ $b=-15$ $c=7$

i) $\alpha\beta = \frac{c}{a}$
 $= \frac{7}{3}$
 $\alpha\beta = 2\frac{1}{3}$ ②

ii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta}$
 $= \frac{-\frac{b}{a}}{\frac{c}{a}}$
 $= \frac{-15}{7} \times \frac{3}{7}$
 $= \frac{-45}{49}$ ②
 $\therefore \frac{1}{\alpha} + \frac{1}{\beta} = 2\frac{1}{7}$

QUESTION 4

(a) i) let $y = x^3 + \frac{1}{x^2}$

$$= x^3 + x^{-2}$$

$$\frac{dy}{dx} = 3x^2 - 2x^{-3}$$

$$\frac{dy}{dx} = 3x^2 - \frac{2}{x^3}$$

ii) let $y = 2xe^{x^2}$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$= e^{x^2} \cdot 2 + 2x \cdot e^{x^2} \cdot 2x$$

$$= 2e^{x^2} + 4x^2 e^{x^2}$$

$$\frac{dy}{dx} = 2e^{x^2}(1 + 2x^2)$$

(ii) let $y = \ln\left(\frac{x}{\cos x}\right)$

$$= \ln x - \ln \cos x$$

$$\frac{dy}{dx} = \frac{1}{x} - \frac{-\sin x}{\cos x}$$

$$\frac{dy}{dx} = \frac{1}{x} + \tan x$$

(b) $\int \sin 2x \, dx$

$$= -\frac{1}{2} \cos 2x + c$$

(c) i) $\int_1^2 \sqrt{3x-2} \, dx$

$$= \int_1^2 (3x-2)^{\frac{1}{2}} \, dx$$

$$= \left[\frac{(3x-2)^{\frac{3}{2}}}{\frac{3}{2} \times 3} \right]_1^2$$

$$= \frac{2}{9} \left[\sqrt{3x-2}^3 \right]_1^2$$

c) cont'd

$$= \frac{2}{9} [8 - 1]$$

$$= \frac{14}{9}$$

$$= 1\frac{5}{9}$$

ii) $\int^e \left(\frac{2}{x} + x\right) dx$

$$= \left[2 \ln x + \frac{x^2}{2} \right]^e$$

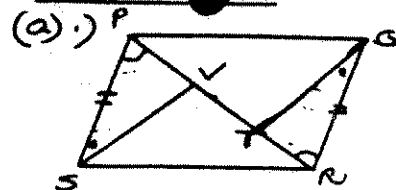
$$= \left(2 \ln e + \frac{e^2}{2} \right) - \left(2 \ln 1 + \frac{1}{2} \right)$$

$$= 2 + \frac{e^2}{2} - 0 - \frac{1}{2}$$

$$= \frac{3}{2} + \frac{e^2}{2}$$

$$= \frac{e^2 + 3}{2}$$

QUESTION 5



i) $\angle PQR = \angle PSR$
(opp. $\angle$'s of $\parallel$ lines are eq.)

ii) In Δ 's PVS & RTQ
 $PS = QR$ (opp. sides of rect)
 $PV = RT$ (diagonals bisect each other)

$\therefore \angle SPR = \angle QRP$
(alt. $\angle$'s, $SP \parallel RQ$)

iii) $\angle PSV = \angle QRT$ (from i)
 $= \angle PSR = \angle QRP$ (from ii)
SV, RT bisect these $\angle$'s (given)

$\therefore \Delta PVS \cong \Delta RTQ$ (A.A.S.)

iv) $PR = 20 \text{ cm}$ & $TR = 8 \text{ cm}$

Now, $TR = PV$ (corresponding sides in congruent Δ 's)

$$\therefore PV = 8 \text{ cm}$$

$\therefore TV = PR - (PV + TR)$
 $= 20 - (8 + 8)$

$$TV = 4 \text{ cm}$$

(b) $P'(x) = 15(5x-1)^2$

$$P(x) = \frac{15(5x-1)^3}{3 \times 5} + C$$

$$10 = (-1)^3 + C$$

$$\therefore C = 11$$

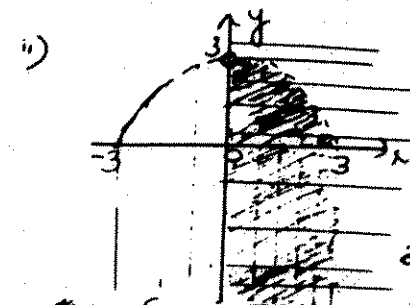
$$P(x) = (5x-1)^3 + 11$$

QUESTION 6

(a) $y = \sqrt{9-x^2}$

i) D: $-3 \leq x \leq 3$

R: $0 \leq y \leq 3$



ii) $\{y < \sqrt{9-x^2} \cap x > 0\}$

iii) $(2, -4)$ lies in region?

sub. in $y < \sqrt{9-x^2}$

$$-4 < \sqrt{9-4}$$

$$-4 < \sqrt{5} \text{ True}$$

sub. in $x > 0$

$$2 > 0 \text{ True}$$

$\therefore (2, -4)$ lies in region 2

$$\{y < \sqrt{9-x^2} \cap x > 0\}$$

(b) i) $P(4, 4) = \frac{1}{4} \times \frac{1}{4}$

$$P(4, 4) = \frac{1}{16}$$

ii) $P(5 \text{ or } 4) = P(1, 3)$

$$+ P(2, 2) + P(3, 1)$$

$$= \left(\frac{1}{4} \times \frac{1}{4} \right) \times 3$$

$$P(5 \text{ or } 4) = \frac{3}{16}$$

QUESTION 6 (cont'd)

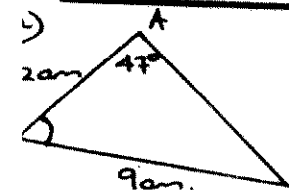
$$\int_0^2 f(t) dt \div \frac{1}{2} [f(0) + f(2)] + 2 [f(1) + f(3) + f(4)]$$

$$\begin{aligned} \text{NS } h &= \frac{b-a}{n} \\ &= \frac{2-0}{4} \\ &= \frac{1}{2} \end{aligned}$$

$$\int_0^2 f(t) dt \div \frac{0.5}{2} [0 + 0.27 + 2(0.3 + 0.37 + 0.3)]$$

$$\int_0^2 f(t) dt = 0.5675 = 0.6 \text{ (1dp)} \quad (3)$$

QUESTION 7:-



$$\frac{\sin B}{b} = \frac{\sin A}{a}$$

$$\frac{\sin B}{12} = \frac{\sin 47^\circ}{9}$$

$$\sin B = \frac{12 \sin 47^\circ}{9}$$

$$\angle B = 77^\circ 12' \text{ (nearest min)} = 77^\circ \text{ (nearest } ^\circ)$$

$$\begin{aligned} \angle ACB &= 180^\circ - (47^\circ + 77^\circ) \\ \angle ACB &= 56^\circ \text{ (nearest } ^\circ) \end{aligned} \quad (2)$$

$$\begin{aligned} A &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 12 \times 9 \times \sin 56^\circ \\ &= 44.76802892 \\ A &= 45 \text{ (2 s.f.)} \end{aligned} \quad (1)$$

$$(b) \text{ AP: } 50, +3, 36, \dots$$

$$a = 50, d = -7$$

$$i) T_n = a + (n-1)d$$

$$T_n = 50 + (n-1)(-7)$$

$$= 50 - 7n + 7$$

$$T_n = 57 - 7n \quad (2)$$

$$ii) T_n = -27 \quad n = ?$$

$$-27 = 57 - 7n$$

$$7n = 57 - 27$$

$$= 84$$

$$n = 12 \quad (2)$$

$\therefore 12$ terms in sequence

$$iii) S_n = \frac{n}{2} (a + l)$$

$$S_{12} = \frac{12}{2} (50 + -27)$$

$$S_{12} = 138 \quad (2)$$

$\therefore$ Sum of series is 138

$$(c) x^2 - (k+3)x + (k+6) = 0$$

$$a = 1; b = -(k+3); c = k+6$$

for equal + rational roots;
 $\Delta \geq 0$

$$b^2 - 4ac \geq 0$$

$$k^2 + 6k + 9 - 4(k+6) \geq 0$$

$$k^2 + 6k + 9 - 4k - 24 \geq 0$$

$$k^2 + 2k - 15 \geq 0$$

$$(k+5)(k-3) \geq 0$$

$$\therefore k = -5 \text{ or } k = 3 \quad (3)$$

QUESTION 8 -

$$(a) y = (x-2)^2(x+1)$$

$$i) \text{ axis } y \text{ axis } \Rightarrow x = 0$$

$$\text{ie } y = (-2)^2 \times 1$$

$$y = 4 \Rightarrow (0, 4) \quad (1)$$

$$\text{axis } x \text{ axis } \Rightarrow y = 0$$

$$\text{ie } (x-2)^2(x+1)$$

$$\text{ie } x = -1 \text{ or } x = 2$$

$$\Rightarrow (-1, 0) + (2, 0) \quad (1)$$

$$ii) \text{ for stationary pts}$$

$$\frac{dy}{dx} = 0$$

$$\text{ie } v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx} = 0$$

$$(x+1) \cdot 2(x-2) + (x-2)^2 \cdot 1 = 0$$

$$(x-2)[2(x+1) + (x-2)] = 0$$

$$(x-2)[3x] = 0$$

$$\therefore x = 2 \text{ or } 0 \quad (2)$$

$$\frac{d^2y}{dx^2} = 3x^2 - 6x$$

$$\frac{d^2y}{dx^2} = 6x - 6$$

$$\text{test } x = 2 \quad \frac{d^2y}{dx^2} > 0$$

$$\therefore \text{min T.P at } (2, 0) \quad (1)$$

$$\text{test } x = 0 \quad \frac{d^2y}{dx^2} < 0$$

$$\therefore \text{max T.P at } (0, 4) \quad (1)$$

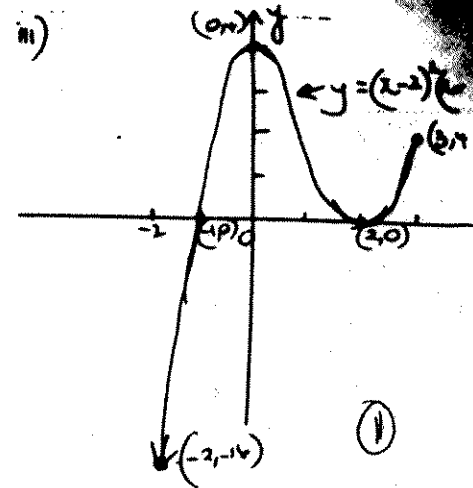
$$iii) \text{ when } x = -2$$

$$y = (-4)^2 \times -1$$

$$= -16 \Rightarrow (-2, -16) \quad (1)$$

$$\text{when } x = 3$$

$$y = 1 \times 4 = 4 \Rightarrow (3, 4) \quad (1)$$



$$(b) y = x^2 + \frac{2}{x} + 4$$

$$= x^2 + 2x^{-1} + 4$$

$$\frac{dy}{dx} = 2x - 2x^{-2}$$

$$= 2x - \frac{2}{x^2} \quad (1)$$

$$\text{at } x = -1$$

$$\frac{dy}{dx} = -2 - \frac{2}{1}$$

$$= -4$$

$$\therefore m = -4 \text{ pt } P(-1, 3) \quad (1)$$

$$m = \frac{y_1 - y_2}{x_1 - x_2}$$

$$-4 = \frac{y - 3}{x + 1}$$

$$y - 3 = -4x - 4 \quad (1)$$

$$4x + y + 1 = 0 \text{ is the eqn of the tangent}$$

QUESTION 9

- (a) $x^2 = 8y$
 $\therefore x^2 = 4ay$
 $\therefore 4a = 8$
 $a = 2$
 i) focal length = $2a$
 ii) Focus $(0, 2)$
 iii) directrix: $y = -2$ ③

(b) $S_0 = ?$ $a = 75 \text{ km}$ $r = \frac{1}{3}$

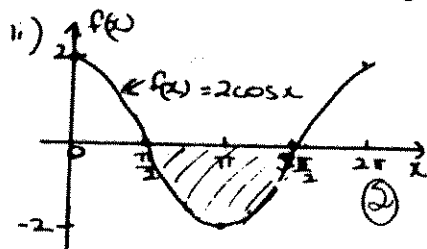
$$S_0 = \frac{a}{1-r}$$

$$= \frac{75}{\frac{2}{3}}$$

$$S_0 = 112 \frac{1}{2} \text{ km.} \quad \textcircled{2}$$

No, she will never reach the watering hole as she can only travel $112 \frac{1}{2} \text{ km}$ which is $< 130 \text{ km}$. ①

(c) $f(x) = 2 \cos x$
 i) Range: $-2 \leq f(x) \leq 2$ ①



$A = \left| \int_{\pi/2}^{3\pi/2} 2 \cos x \, dx \right|$
 $= \left| \left[2 \sin x \right]_{\pi/2}^{3\pi/2} \right|$
 $= \left| (2(-1)) - (2(1)) \right|$
 $= \left| -4 - 2 \right|$
 $A = 6 \text{ u}^2$ ③

QUESTION 10

- (a)  i) $P_{\text{sector}} = 8 \text{ cm}$
 Arc length $AB = r\theta$
 $\therefore \text{Perimeter} = r\theta + 2r$
 $\therefore 8 = r\theta + 2r$
 $\frac{8}{r} = \theta + 2$
 $\theta = \frac{8}{r} - 2$ (as req.) ②

Area = $\frac{1}{2} r^2 \theta$
 $= \frac{1}{2} r^2 \left(\frac{8}{r} - 2 \right)$
 $= \frac{1}{2} \times 8r - r^2$
 $A = 4r - r^2$ (as req.)

ii) $\frac{dA}{dr} = 4 - 2r$ ①
 $\frac{d^2A}{dr^2} = -2$ ②
 For stationary values $\frac{dA}{dr} = 0$
 $\therefore 4 - 2r = 0$
 $2r = 4$
 $r = 2$ ①

test $r = 2$ $\frac{d^2A}{dr^2} = -2 < 0$

$\therefore$ max value when $r = 2$

iii) Max $A = 4 \times 2 - 2 \times 2$
 $= 8 - 4$
 $\text{max Area} = 4 \text{ cm}^2$ ①

QUESTION 10 (cont'd)

(b) $P = \$2300$ $n = 25, 24, 23, \dots, 1$
 $r = 1.12$

Let A_n be the amt the investment grows to after n investments

$A_1 = 2300 (1.12)^{25}$

$A_2 = 2300 (1.12)^{24}$

$A_3 = 2300 (1.12)^{23}$

$A_{24} = 2300 (1.12)^2$

$A_{25} = 2300 (1.12)^1$

Total Amt = $A_{25} + A_{24} + \dots + A_1$
 $= 2300 (1.12 + 1.12^2 + 1.12^3 + \dots + 1.12^{25})$
 G.P. $a = 1.12$, $n = 25$, $r = 1.12$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$S_{25} = \frac{1.12 (1.12^{25} - 1)}{0.12}$

Total Amt = $2300 \times S_{25}$
 $= 343468.0491$
 $= \$343468.05$ (nearest $\text{\$}$)

$\therefore$ The investment is worth $\$343468.05$ (nearest $\text{\$}$) after 25 years

⑤